$AM = MB$ and so, $x - x_1 = x_2 - x$

and this is simplified to $x = \dfrac{x_1 + x_2}{2}$

Now it is easy to discuss the general case. If $A(x_1, y_1)$, $B(x_2, y_2)$ are any two points and $M(x, y)$ is the mid-point of the line segment AB, then M' is the mid-point of AC (in the Fig. 5.22). In a right triangle the perpendicular bisectors of the sides intersect at the mid-point of the hypotenuse. (Also, this property is due to similarity among the two coloured triangles shown; In such triangles, the corresponding sides will be proportional).

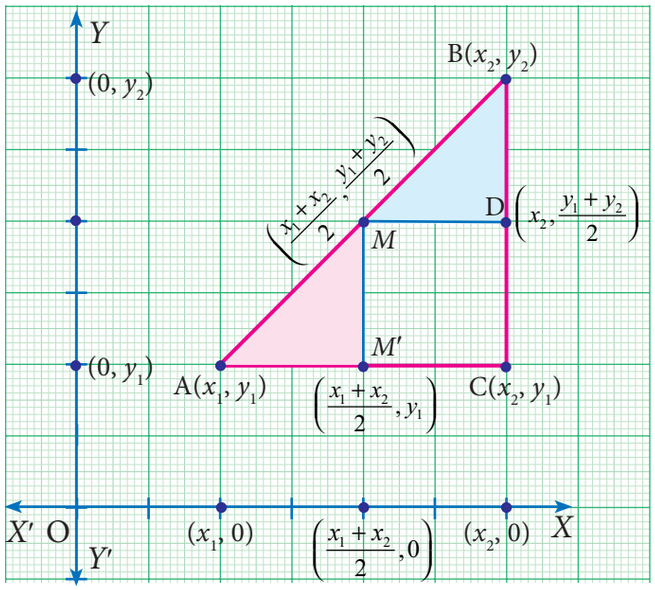


Fig. 5.22

Another way of solving

(Using similarity property)

Let us take the point M as $M(x,y)$

Now, $\Delta AMM'$ and ΔMBD are similar. Therefore,

$$\frac{AM'}{MD} = \frac{MM'}{BD} = \frac{AM}{MB}$$

$$\frac{x - x_1}{x_2 - x} = \frac{y - y_1}{y_2 - y} = \frac{1}{1} \quad (AM = MB)$$

Consider, $\dfrac{x - x_1}{x_2 - x} = 1$

$2x = x_2 + x_1 \Rightarrow x = \dfrac{x_2 + x_1}{2}$

Similarly, $y = \dfrac{y_2 + y_1}{2}$

x-coordinate of M = the average of x-coordinates of A and C = $\dfrac{x_1 + x_2}{2}$ and similarly,

y-coordinate of M = the average of y-coordinates of B and C = $\dfrac{y_1 + y_2}{2}$

The mid-point M of the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Thinking Corner

If D is the mid-point of AC and C is the mid-point of AB, then find the length of AB if $AD = 4cm$.

For example, The mid-point of the line segment joining the points $(-8, -10)$ and $(4, -2)$ is given by $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$ where $x_1 = -8$, $x_2 = 4$,

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